

4 Bit Binary Adder

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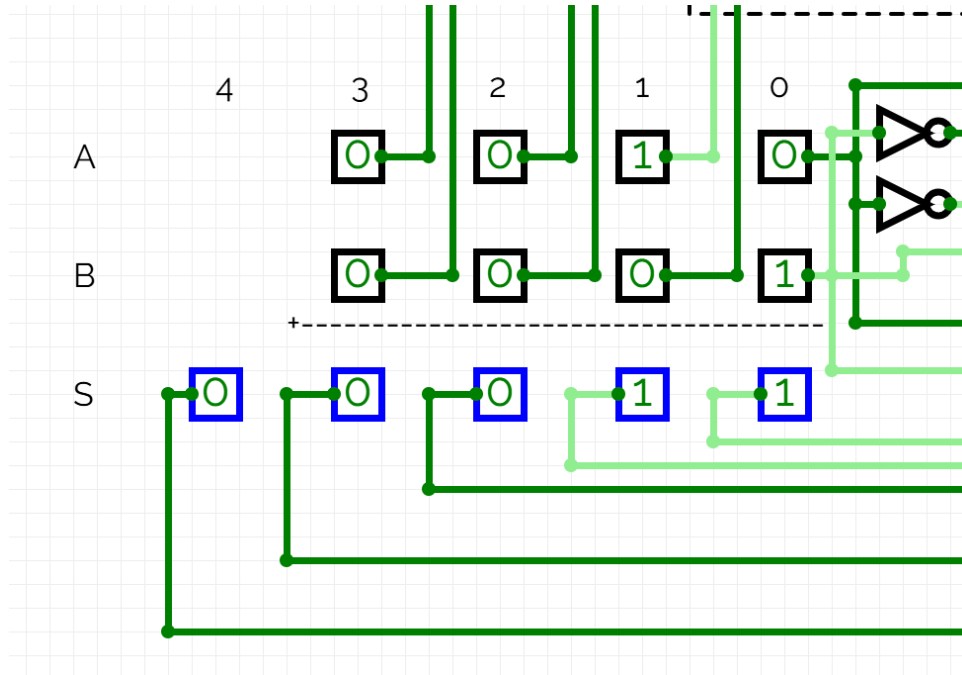
UAT

RBT131: Digital Logic Basic Processor Design

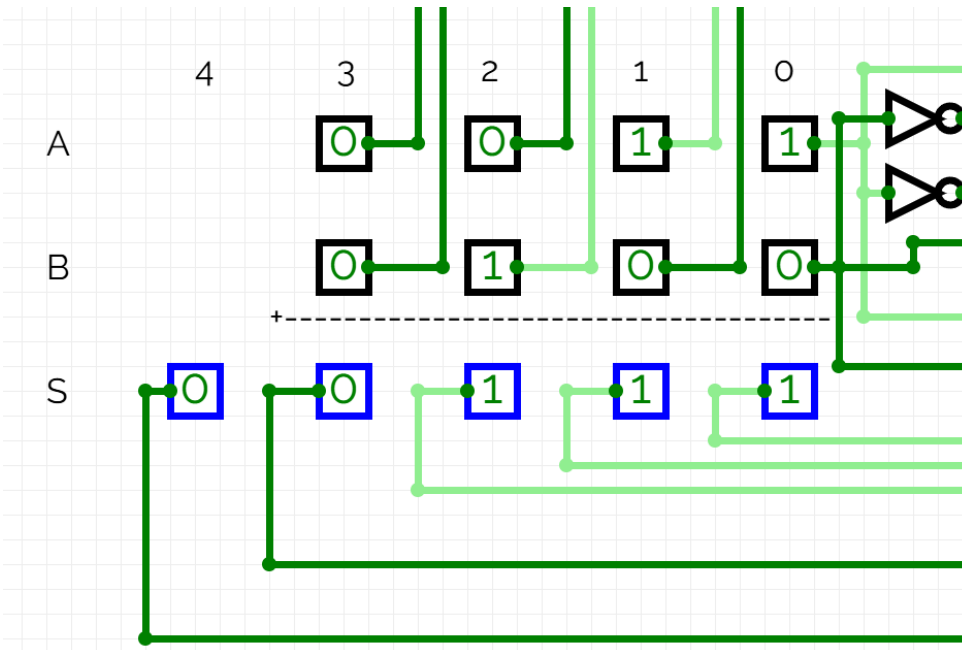
Prof. Prater

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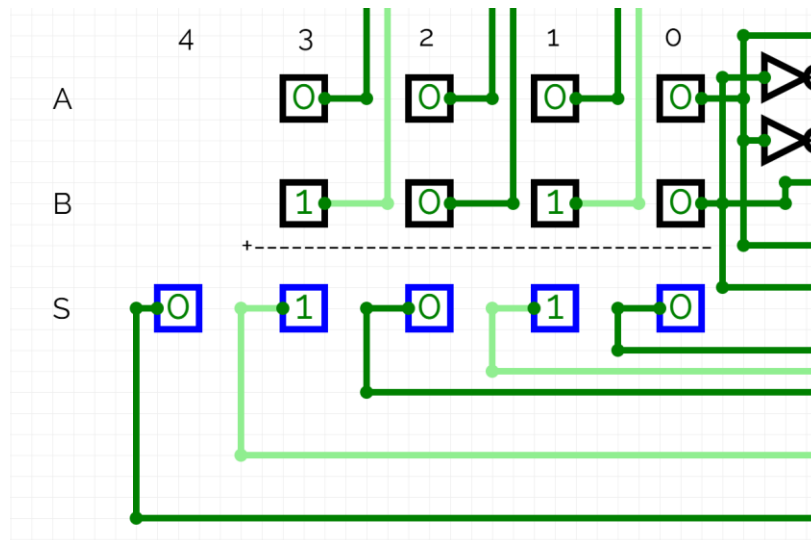
Examples



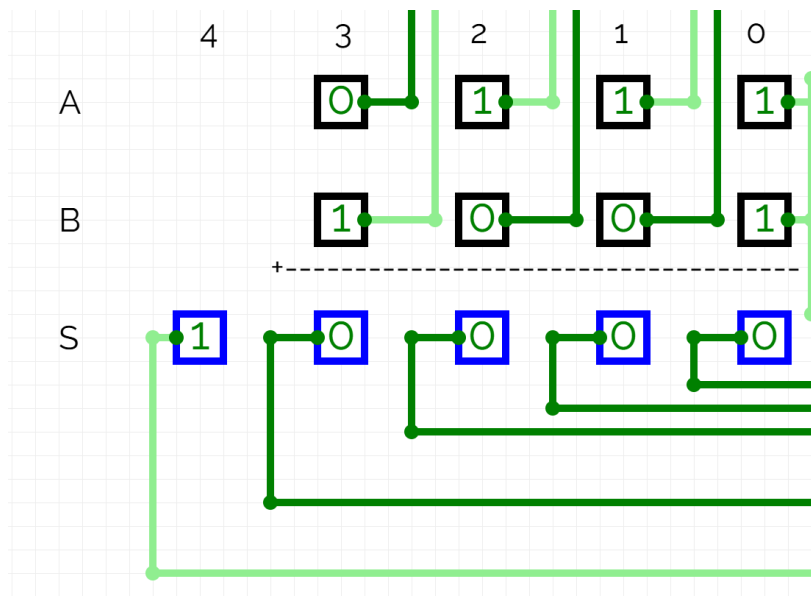
A = 0010, B=0001, S=00011



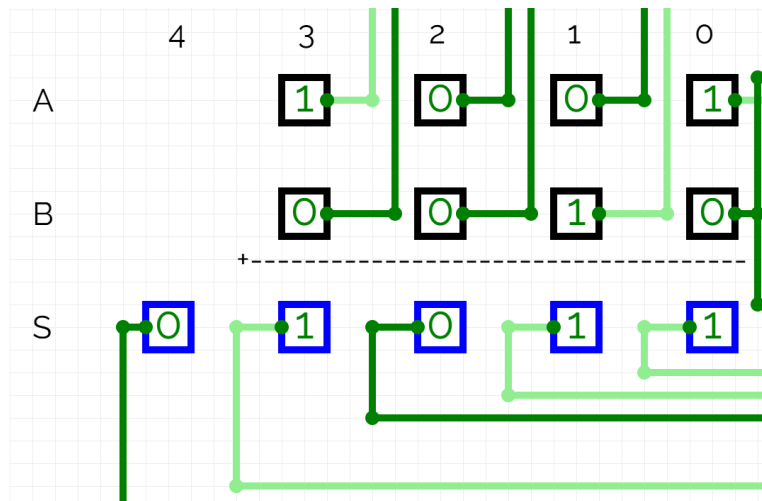
A=0011, B=0100, S=00111



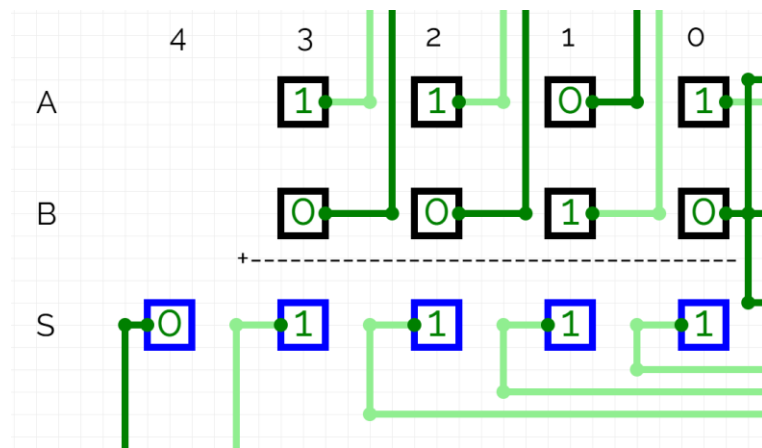
A = 0000, B = 1010, S = 01010



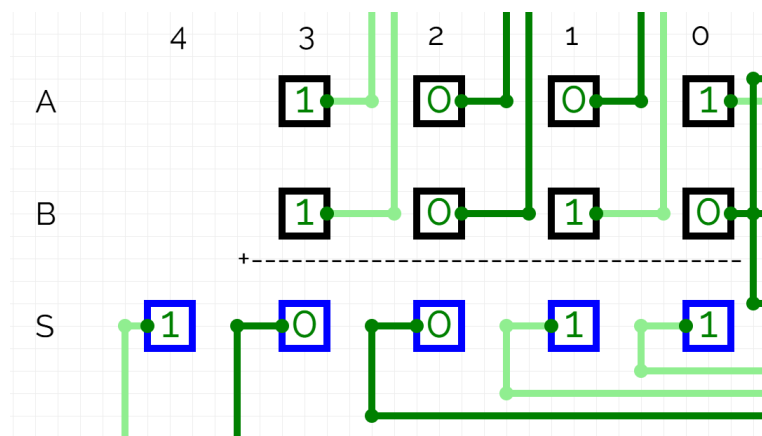
A=0111, B = 1001, S = 10000



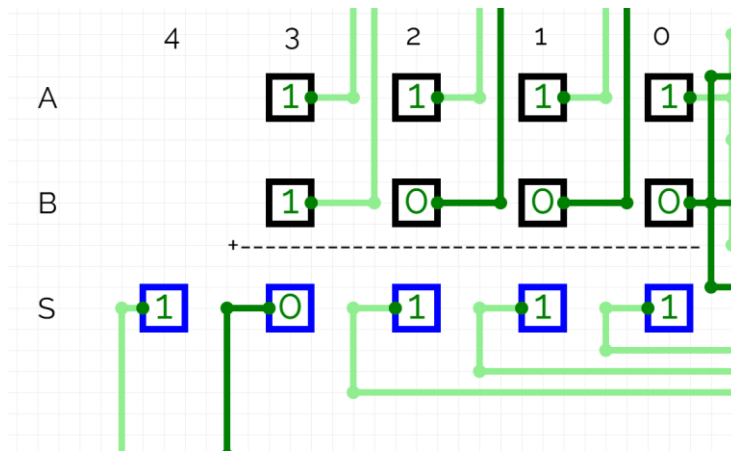
A = 1001, B = 0010, S = 01011



A = 1101, B = 0010, S = 01111



A = 1001, B = 1010, S = 10011



A=1111, B=1000, S=10111

Explanation

A 4-bit binary adder works by combining several full adders to add 2 4-bit binary values (A and B), this results in a 5-bit result in case the final bit overflows and needs a carry. Each full adder takes in the result from the previous adder along with the carry result (if necessary). The math applies as follows:

- zero 1s = 0
- one 1 = 1
- two 1s = sum 0 and carry 1
- three 1s = sum 1 and carry 1

This process can easily be repeated for adding together numbers with any number of bits.